

1 Graphic Machinery - Layout Editor

Overview

Menu	Production facility management → Current information → Graphic machinery Human resources management → Access control → Security control center Information management → Production overview → Graphic machinery
Transaction code	Mpark
Function authorization	mparkle.layedit
Function key	 Layout editor

Purpose

Use the layout editor of the [Graphic machinery](#) to "design" layouts showing user-defined organizational units (e.g. halls, departments, etc.).

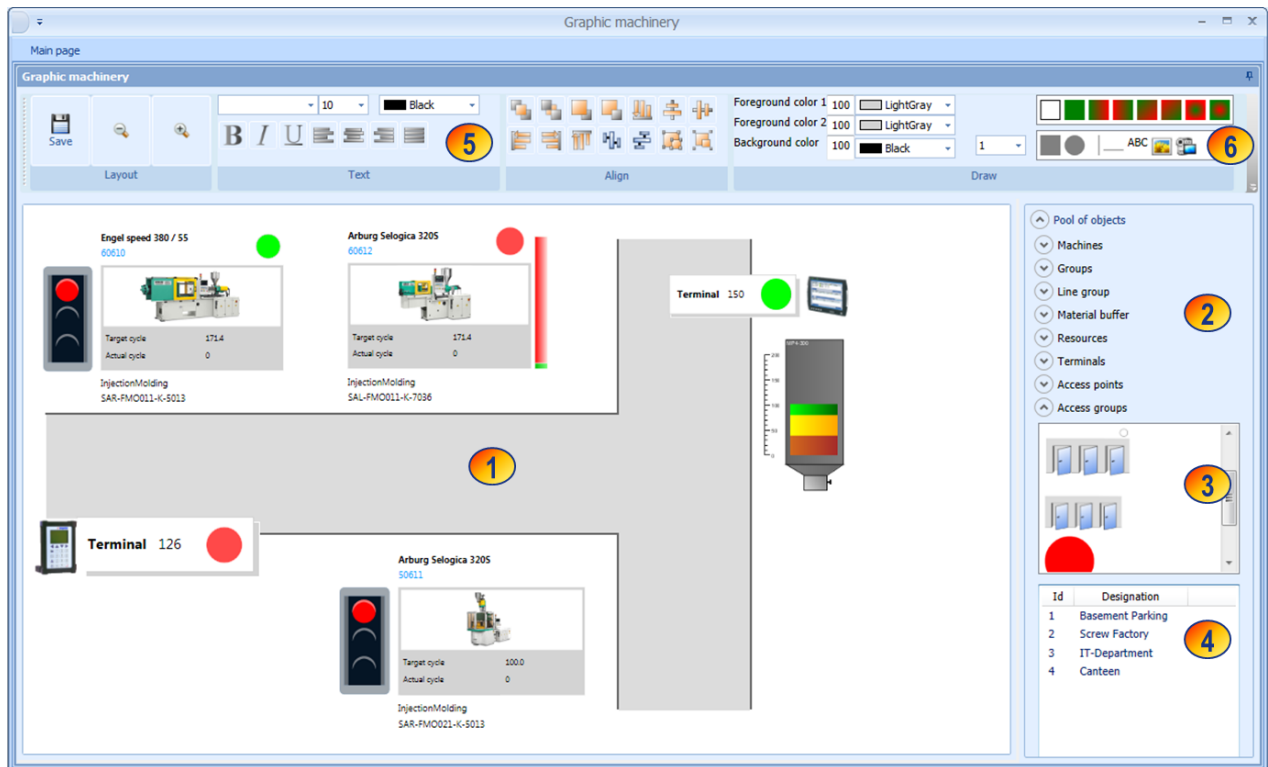
Integration

The layout editor is integrated in the application [Graphic machinery](#). Use the button  to start the layout editor.

Requirements

The objects that you want to show in a layout must be available in the system master data. If necessary, you must create the objects in the master data. And you must have created a layout entry for the [Graphic machinery](#).

1.1 Layout editor



- (1) Layout panel
- (2) Pool of objects: If you click an object type (e.g. machine), the dialog shows the following additional data:
- (3) Templates of the object type
- (4) Available objects (e.g. machines)
- (5) The upper panel shows the "tools" you can use to edit a layout graphically (this is an example; chapter [Toolbar](#) describes all available icons).
- (6) Graphic objects: templates, rectangle, circle, lines, text, image, video and/or video stream (mms).



In the layout editor you cannot start the help via F1. Start the help by clicking the button *Help on application.*

Toolbar



Please note that the layout editor does not provide a quick launch bar and you cannot minimize the toolbar.

The layout editor does not support functions of the MES Development Suite.

The layout editor does not support the functions  "print preview" and  "print all".

Layout category



Save

Click "save" if you changed the layout.



Note: No confirmation prompt appears when exiting the application, even though changes have been made.



Zooms out the layout in the layout editor.

Note: Changing the size in the layout editor does not affect the data displayed in the graphic machinery.



Zooms in the layout in the layout editor.

Note: Changing the size in the layout editor does not affect the data displayed in the graphic machinery.

Text category

You can use the functions described below to format a static text that you integrated in the layout using

drag and drop via the button .



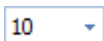
Changes always affect the entire text within a text field.

Note:

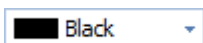
- the buttons do not show current formatting of a text and
- font type, font size and color are also reset if you change the format (e.g. change to *italics*).



Changes the font of a static text.



Changes the font size of a static text.



Changes the font color of a static text. Changes the entire text included in the text field.



Shows the text in bold letters.

Click the button once more to undo formatting.



Shows the text in italic letters.

Click the button once more to undo formatting.



Underlines the text.

Click the button once more to undo formatting.



Left aligns static text:

asasjaskj sjd aksjda ksjdlaksdj
alksjdalksjda lkdjklaksöa jdlaksjd
aksjdak lsdja iedkpo qwie qowiep
qoiweq poeiqow oqw pow iqpo
weipoq eipqowi epqowieqoi



Center aligns static text:

asasjaskj sjd aksjda ksjdlaksdj
alksjdalksjda lkdjklaksöa jdlaksjd
aksjdak lsdja iedkpo qwie qowiep
qoiweq poeiqow oqw pow iqpo
weipoq eipqowi epqowieqoi



Right aligns static text:

asasjaskj sjd aksjda ksjdlaksdj
alksjdalksjda lkdjklaksöa jdlaksjd
aksjdak lsdja iedkpo qwie qowiep
qoiweq poeiqow oqw pow iqpo
weipoq eipqowi epqowieqoi



Justifies static text:

asasjaskj sjd aksjda ksjdlaksdj
alksjdalksjda lkdjklaksöa jdlaksjd
aksjdak lsdja iedkpo qwie qowiep
qoiweq poeiqow oqw pow iqpo
weipoq eipqowi epqowieqoi

Alignment category

Use the functions of this category to align graphic objects.



Brings the currently selected item to the foreground



Brings the currently selected item into the background



Moves the currently selected item one level forward.



Moves the currently selected item one level back.



Left aligns the selected items.



Right aligns the selected items.



Top aligns the selected items.



Bottom aligns the selected items.

Draw category

Use the functions described below to format a graphic object that you integrated in the layout using drag and drop via the buttons   .



Changes always refer to all selected objects. Note:

- The buttons do not show current formatting of a graphic element and
- the color is also reset if the format is changed (e.g. changed color gradient direction).

Background color

100 LightGray



Changes the object color for objects without color gradient.

Changes the first color for objects with color gradient.

Changes the background color for text fields; you can change the foreground color by choosing a color in the *text* category.

2nd background color

100 LightGray



Changes the second color of objects with color gradient.

With text fields, the background color changes; you can change the foreground color by choosing the color in the *text* category.

Line color/border color

100 Black



Changes the line color.



Changes the border color of the object.

1



Changes the line thickness.



Specifies the fill effect, i.e. fill or gradient.

With text, the background is specified.

Left-click to select a fill effect before drawing one of the objects - .



Click this icon if you want to draw a rectangle (rectangle, square). Drag and drop the item to the layout.



Click this icon if you want to draw a circle or oval. Drag and drop the item to the layout.



Click this icon if you want to draw a vertical line. Drag and drop the line in the defined thickness (see above) to the layout.

Once you have dropped the icon in the layout panel, a square is shown. You can change the size and the shape of the object as long as it is selected.

Note: You cannot draw diagonal lines.



Click this icon if you want to draw a horizontal line. Drag and drop the line in the defined thickness (see above) to the layout.

Once you have dropped the icon in the layout panel, a circle is shown. You can change the size and the shape of the object as long as it is selected.

Note: You cannot draw diagonal lines.



Click this icon if you want to enter a (static) text. Drag and drop the text in the defined font, font size, color and/or background color (see above) to the layout.

Once you have dropped the icon in the layout panel, the application displays a text field showing the letters "ABC". You can change this text as long as the object is selected.



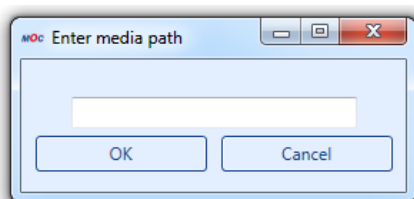
Use this icon to insert an image into the layout. Once you have dropped the icon in the layout panel, a "File Open" dialog appears where you can select and insert the image into the layout.

The following formats are supported: jpg, png, bmp, gif.

Note: The image is transferred to the layout, i.e. to the zip file of the layout. Therefore, the file size affects the load time of the layout. We recommend using compressed file formats for images. The file size should range between double-digit and triple-digit kilobytes at most.



Use this icon to integrate videos or video streams. Select the icon, drag it into the layout and then drop it. A dialog opens where you can enter the path of the video or video stream.



The following formats are supported:

- Video: mp4, wmv

You can enter an absolute or relative path referring to the MOC directory.

- Video stream: mms

The MMS protocol (Microsoft Media Server Protocol) is a protocol developed by Microsoft that has been designed to transfer multimedia streams. The stream has to start with mms://, e.g. mms://192.168.50.10:8080/

Note: The video itself is not integrated in the layout, but the path to the video. Therefore, make sure that the video is stored in a central location that all MOC clients can access.



Note: If you select an object in a layout, the functions provided are adjusted to the properties of the selected object. Except for these options:

- Underlined
- Alignment (right, left, center, justify)

Operation



Note that the layout editor does not provide an "undo" function.

Integrating a graphic object, e.g. rectangle, line, text, image, etc. in the layout

In general, you can position a graphic object in the layout panel as follows:

1. If you want to add a rectangle, circle or text, choose the fill effect at first by left clicking:



. The selected icon is highlighted in color.

2. Left-click to select an object: . The selected icon is highlighted in color.

3. Hold the left mouse button down and drag this object to the layout panel and drop it at the required position (drag&drop).

4. If you insert an image or video/video stream, a dialog opens to select the image or an input field opens where you can enter the path of the video and/or video stream.

5. As long as the object is selected,



you can change its properties or move the object to another position by drag&drop.

Selecting an object in the layout

To select a single graphic object, click that object. A "border" is shown around the object. Please find below an example for a square:

Before (before clicking):



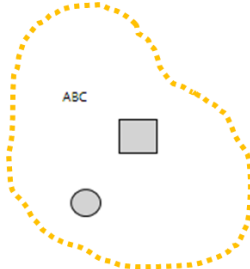
After (after clicking):



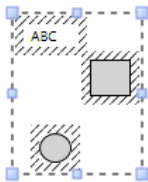
Selecting several objects in the layout

To select multiple objects at a time, hold the left mouse button down and "go" around the required objects so that an orange-colored, dotted circle occurs. Then release the left mouse button:

Before releasing the left mouse button:



After releasing the left mouse button:



Changes to the format, such as changing the (background) color now affect all selected objects.

Integrating objects, e.g. machine, group, etc. in the layout

Proceed as described below to display an object, e.g. a machine, group, in the layout:

1. Make sure the right detail panel shows the object types. To do so, click the text "pool of objects". Then all object types are displayed (the number is only displayed in the above [screenshot](#); the number is only for orientation and is not shown in the application):

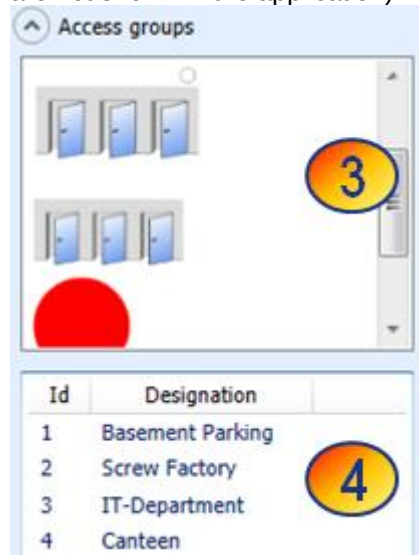


- Now click the object type you want to insert in the layout. Two windows appear below the object type:

Top (3): the templates that are available for this object type

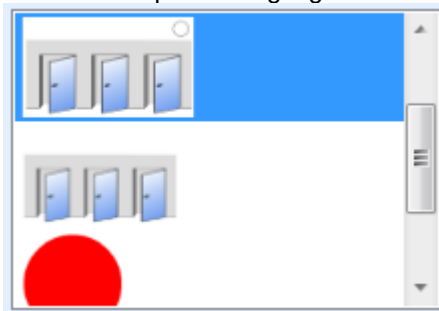
Bottom (4): the objects of this type available in HYDRA

(the numbers are only displayed in the above [screenshot](#); the numbers are only for orientation and are not shown in the application):

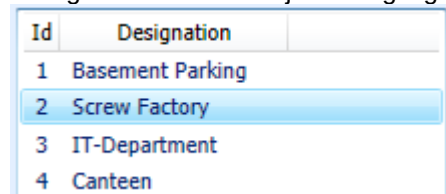


Please wait a few seconds until the below window shows IDs and designations. The number of columns and column headers in the below selection dialog depends on the object type.

- Now select the template you want to show for the object in the layout by left clicking the template. The selected template is highlighted in color:



- Now select the object you want to show in the layout by left clicking the object in the lower selection dialog. The selected object is highlighted in color:



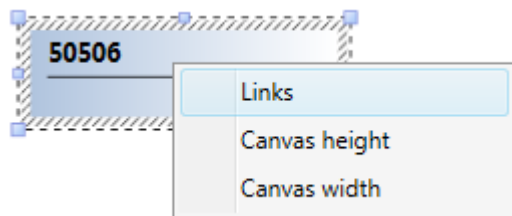
5. Hold the left mouse button down and drag this object from the selection window to the layout panel and drop it at the required position (drag&drop).
6. Click the object in the layout panel to change its size or to move it to another position by drag & drop.



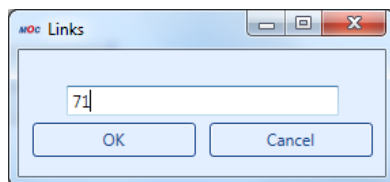
Inserting links between layouts

Use the context menu to add a link to another layout for objects. To do so, enter the appropriate layout ID. Proceed as follows:

1. Right-click the object you want to link. It does not matter if it is an inserted layout or another object.
2. Select the context menu item: Links



3. Enter the layout ID you want to add.



4. Confirm with ok.

Note: You can find the layout ID for the link in the list of layouts in the *Graphic machinery* (shop floor monitor). If the column indicating the layout ID is not displayed, add the column to the list by choosing the option "select columns".

Adjusting drawing canvas

Use the context menu of the layout editor to adjust the available drawing canvas. In the context menu, select the option "canvas height" and "canvas width" and enter the required values.

Templates in the Graphic Machinery

The graphic machinery provides ready-made templates for the below-mentioned object types.

- Workplaces/machines

- Groups (capacity groups)
- Lines (workplaces/machines of the "L" type)
- Material buffer
- Resources (no DNC and file resources)
- Terminals
- Access points
- Access groups
- KPIs of energy resources
- Machine-specific energy meters
- Key performance indicators (current, shift-related KPIs for workplaces/machines)
- Operations logged on (up to three operations)



You can only add and show KPIs if you enable the extensions [mpark2kpi](#).



You can only add and show the operations logged on if you enable the extensions [mpark2roptemp](#).

You can change existing templates as part of a customization.

Deployment of layouts

You can pass created layouts from one system to the other. MPDV explains how to pass layouts as part of a service.

1.2 Notes on tooltip templates

Workplaces/machines

The tooltip shows the following data:

Workplace and machine data:

- Workplace – Unique identification of the workplace/machine according to the [Configuration](#).
- Designation – Name of the workplace according to the [Configuration](#).
- Workplace category – Workplace category according to the [Configuration](#).
- Status – Status text of the status currently set for the workplace
- Status since – Beginning of the current status
- Target cycle – Current target cycle (format: hours:minutes:seconds per 1000 cycles)
- Actual cycle – Current actual cycle (format: hours:minutes:seconds per 1000 cycles)
- Yield (P) – Yield recorded for the workplace in the current shift (primary quantity unit)

- Scrap (P) – Scrap recorded for the workplace in the current shift (primary quantity unit)

Operation related data for the currently logged on operation. If several operations are logged on to the workplace at the same time, the data of the operation last logged on is displayed.

- MES order number
- OP designation
- Article – Article defined in the operation
- Article designation – Article designation defined in the order
- Tool – Tool defined for the operation
- Yield (P) – Yield posted for the operation (primary quantity unit)
- Scrap (P) – Scrap posted for the operation (primary quantity unit)
- Open quantity (P) – Open quantity posted for the operation (primary quantity unit)
- Rework (P) – Rework quantity posted for the operation (primary quantity unit)

Groups

The tooltip shows the following data:

- Group – Identifies the group according to the [Configuration](#)
- Designation – Name of the group according to the [Configuration](#)
- Cost center – Cost center of the group according to the [Configuration](#)

Line group

The tooltip shows the following data for line workplaces (type L):

- Workplace – Unique identification of the workplace/machine according to the [Configuration](#).
- Designation – Name of the workplace according to the [Configuration](#).
- Workplace category – Workplace category according to the [Configuration](#).

Terminal

Terminal templates do not provide any tooltips.

Material buffer

The tooltip shows the following data:

- Material buffer – Identifies the material buffer according to the [Configuration](#)
- Material – Material number according to the [Configuration](#)

Resource

The templates for resources show the resource status.

- Resource – Identifies the resource according to the [Configuration](#)
- Resource type – Resource type of the resource according to the [Configuration](#)
- Designation – Name of the resource according to the [Configuration](#)
- Current storage location – Current storage location of the resource
- Resource status – Number of the current resource status
- Resource status – Name of the current resource status according to the [Configuration](#)

Access points

The tooltip shows the following data:

- Access point – Number of the access point according to the [Configuration](#)
- Designation – Name of the access point according to the [Configuration](#)

Access groups

The tooltip shows the following data:

- Access group – Number of the access group according to the [Configuration](#)
- Designation – Name of the access group according to the [Configuration](#)
- Location – Location of the access group according to the [Configuration](#)
- Number status OK – Number of access points with status OK
- Number status error – Number of access points with status "error"
- Number status not active – Number of access points with status "not active"
- Status color access group – for internal use

KPIs of energy resources

The templates for KPIs of energy resources do not provide any tooltips.

Machine-specific energy meters

The templates for machine-specific energy meters do not provide any tooltips.

Key performance indicators (current, shift-related KPIs for workplaces/machines)



You can only add and show KPIs if you enable the extensions [mpark2kpi](#).

You can choose from the following key performance indicators:

KPI	Formula ID used for changes in the formula management and for the configuration of limit values (optional)
Rate of capacity utilization	rcu
Assignment utilization rate	ocu
Techn. efficiency	tec_ef
Rate	yie_ra
Scrap rate	scr_ra
OEE	oee
Availability	avail
Performance	pf_rat
Quality	qual
Machine run time	mch_rt
Actual utilization	act_ut
Yield utilization	yie_ut

Refer to the following document [MDE_KPI_Configuration.pdf](#) for further information on the configuration.



The templates for KPIs do not provide any tooltips. Create a static text and enter the workplace. This way, later on you will know the workplace this KPI refers to.

Operations logged on



You can only add and show the operations logged on if you enable the extensions [mpark2roptemp](#).

The template shows up to three logged on operations including the following information on the operation:

- Article number
- Target quantity (P)
- Yield (P)

- Scrap (P)

- Scrap ratio in %

The system calculates the relation of scrap (P) to the total yield quantity (P) and scrap quantity (P). The result includes two decimal places. The application indicates "undef." if the scrap rate cannot be calculated.

The values are displayed left aligned. The point in time of the operation logon specifies the order in which the application shows the data of the logged on operations. The operation logged on last is displayed first.

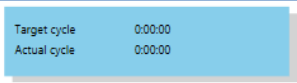
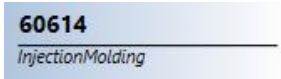
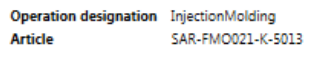


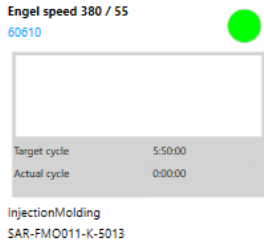
The template does not provide tooltips. Create a static text where you can enter the workplace. Therefore, you can identify later to which workplace this information refers.

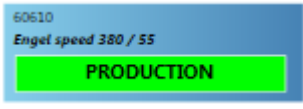
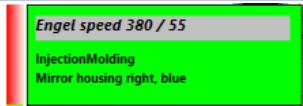
1.3 Description of the templates for workplaces/machines

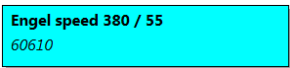
⚙ Available as of service pack 13/2018:

In the [Status assignment](#), you can specify a time for each machine status in field "Warning in the Graphic Machinery". If a machine status lasts longer than the specified time, the machine status symbol starts to flash. If the template supports the flashing status symbol, the machine status is marked with ⚙.

Template name	Screenshot	Description and content of the template
Machines_Cycles		Display of target and actual cycle for the workplace/machine. For the display, the format is used that is defined for the display in the MOC.
Machines_Description		Display of the workplace/machine number and of the name.
Machines_OperationArticle		Display of OP name and of article of the logged on operation. If several operations are logged on, the operation logged on last is displayed.

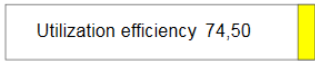
Template name	Screenshot	Description and content of the template
Machines_QuantityLevel		Graphic comparison of the target quantity and the yield produced up to now (in primary quantity unit) of the currently logged on operation. If several operations are logged on, the information of the operation logged on last is displayed.
Machines_Template1		Display of the following information: - Machine name - Machine number - Graphic display of machine status ⚙ - Current target cycle and actual cycle of the machine in the format that has been defined for the display in the MOC. - Operation designation - Article name of the operation You can use the free space to add an image of the respective machine.
Machines_Template2		Display of the following information: - Machine number - Machine name - Graphic and text display of machine status ⚙ - Primary target quantity, yield and scrap of the operation logged on. - Current target cycle and actual cycle of the machine in the format that has been defined for the display in the MOC.

Template name	Screenshot	Description and content of the template
		- Operation designation - Article name of the operation - Graphic comparison of the target quantity and the yield produced up to now (primary quantity unit) of the currently logged on operation.
Machines_Template3		Display of the following information: - Machine number - Machine name - Graphic and text display of machine status ⚙
Machines_Template4		Display of the following information: - Machine name - Operation and article name of the logged on operation - Color display of machine status as background color ⚙ - Graphic comparison of the target quantity and the yield produced up to now (primary quantity unit) of the currently logged on operation.
Machines_Template5		Display of the following information: - Machine name - Machine number - Operation and article name of the logged on operation - Current target and actual cycle of the machine using the unit that has been defined for the display in the MOC.

Template name	Screenshot	Description and content of the template
		- Graphic comparison of the target quantity and the yield produced up to now (primary quantity unit) of the currently logged on operation.
Machines_Template6		Display of the following information: - Machine number - Machine name - Color display of machine status as background color ⚙
Machines_TrafficLight		Traffic light display including the following information ⚙: - Color red: Status "Not assigned" - Color yellow: All statuses except "Production" - Color green: Status "Production"
Machines_StatusCircle		Color display of machine status ⚙ using the colors defined with the status text.
Machines_MaintenanceStatus1 Available as of service pack 13/2018		If the activity/maintenance calendar includes at least one maintenance that is due, this maintenance due is displayed using a symbol. If there is at least one "red" maintenance due, the background color is red. If there is at least one "yellow" maintenance due, the background color is yellow. If there is at least one "blue"

Template name	Screenshot	Description and content of the template
		maintenance due, the background color is blue. If no maintenance is due, no symbol and no background color is displayed.
Machines_MaintenanceStatus2 Available as of service pack 13/2018		See Machines_MaintenanceStatus1

1.4 Description of the templates for KPIs

Template name	Screenshot	Description and content of the template
keyfigure_template_label		Display of the following information: - KPI name - current shift-related KPIs - color display of the KPI as background color
keyfigure_template_label_border		Display of the following information: - KPI name - current shift-related KPIs - color display of the KPI as bar
keyfigure_template_simple		Display of the following information: - current shift-related KPIs - color display of the KPI as background color